

GUSTAVO MARCELO NUÑEZ

PhD Candidate in Computer Science • Knowledge Graphs • GraphRAG • Marine AI

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PROFESSIONAL PROFILE

Computer scientist and PhD candidate working at the intersection of neuro-symbolic artificial intelligence, knowledge graphs, large language models, and marine informatics. Research focuses on integrating structured marine knowledge with GraphRAG and multi-agent architectures to support grounded reasoning, explainable retrieval, and natural-language access to heterogeneous biodiversity and oceanographic data. Combines peer-reviewed research, open-source prototyping, software development, and university teaching experience.

RESEARCH INTERESTS

Knowledge Graphs • GraphRAG and RAG • Neuro-symbolic AI • LLM Evaluation • Semantic Web • Multi-agent Systems • Linked Open Data • Marine Biodiversity and Oceanographic Data

EDUCATION

PhD in Computer Science | Universidad Nacional del Sur (UNS) | Oct 2024–Present

Thesis: Neuro-symbolic artificial intelligence for the exploitation of oceanographic data.

Supervisors: Dr. Pablo R. Fillottrani and Dr. Marcos D. Zárate; research conducted at CESIMAR-CONICET, Puerto Madryn.

Current focus: graph-based retrieval, ontology-grounded reasoning, semantic soundness, and grounding fidelity for LLM-generated answers.

BSc in Computer Science (Licenciatura en Informática) | UNPSJB | Mar 2018–Mar 2024

Undergraduate thesis: Visualizations of Linked Open Data for Marine Sciences.

University Programming Analyst | UNPSJB | 2018–2021

RESEARCH EXPERIENCE

Doctoral Researcher | CESIMAR-CENPAT-CONICET | Aug 2024–Present

Designing neuro-symbolic methods that combine marine knowledge graphs, LLMs, and structured graph retrieval.

Developing evaluation methods for semantic soundness and grounding fidelity in RAG/GraphRAG systems.

Doctoral Consortium paper accepted at the 25th International Semantic Web Conference (ISWC 2026).

Research Project Member | PICT-2021-I-INVI-00245 | Feb 2023–Feb 2025

Contributed to knowledge extraction, reasoning, visualization, and FAIR publication of Argentine marine-science data.

Collaboration among CESIMAR-CONICET, LINVI-UNPSJB, GILIA-UNCOMA, and CIT-GSJ.

Research Project Member | Southern elephant seal computer vision projects | 2023–2025

Applied convolutional neural networks to remote detection, classification, and counting from digital imagery for conservation monitoring.

CIN Research Fellow | UNPSJB | Sep 2022–Sep 2023

Project: Visual Analytics for Linked Open Data Applied to Marine Sciences.

SELECTED RESEARCH AND SOFTWARE PROJECTS

HERMES — GraphRAG evaluation platform | Python, Streamlit, NetworkX, GraphRAG, OpenAI, Mistral

Generates and compares six knowledge graphs and multiple LLMs over a shared historical marine-biology corpus; supports traceable Q&A and post-hoc symbolic evaluation.

AquaMind — Multi-agent marine knowledge system | Python, LLMs, WoRMS API, OBIS API, geospatial data

Integrates WoRMS, OBIS, and marine ecoregion sources through natural-language interaction, semantic reasoning, and coordinated agents.

OBISBot / BotGBIF — Conversational biodiversity access | Python, OBIS API, GBIF API, NLP, LLMs

Makes global marine and biodiversity occurrence records accessible through natural-language queries without requiring API expertise.

ODP-Dashboard — Linked Open Data visualization | Angular, SPARQL, Linked Open Data
Integrates heterogeneous South Atlantic marine-conservation data and visual analytics for scientific exploration.

Mapyzer — Spatio-temporal visualization | Angular, Leaflet, GeoJSON
Transforms georeferenced datasets into interactive exploration-ready maps.

SELECTED PUBLICATIONS AND SCIENTIFIC CONTRIBUTIONS

Doctoral research on neuro-symbolic marine knowledge systems. ISWC 2026 Doctoral Consortium — accepted.

Digitizing and Structuring Early Marine Biodiversity Records: A GraphRAG-Based Methodology. 1st GOBLIN Workshop on Knowledge Graph Technologies, Leipzig, 2025. [View](#)

Integración de Grafos de Conocimiento y Modelos RAG para la Gestión Inteligente de Datos Oceanográficos. WICC 2025, Mendoza, Argentina. [View](#)

ODP-DASHBOARD: Enhancing Marine Species Conservation in the South Atlantic through Linked Open Data Integration. Journal of Computer Science & Technology, 24(2), 2024. [View](#)

Modelo de Redes Neuronales Convolucionales para la Detección, Clasificación y Conteo de Elefantes Marinos a Partir de Imágenes Digitales. WICC 2024, Puerto Madryn, Argentina. [View](#)

Integrating Oceanographic Sensor Data Using SSN/SOSA Ontology. XI Cloud Computing, Big Data & Emerging Topics, 2023. [View](#)

Knowledge Graphs para accesibilidad y reutilización de activos digitales del Mar Argentino. WICC 2023, Junín, Argentina. [View](#)

Democratizing Argentine Marine Science Data Through Linked Open Data. CACIC 2022, La Rioja, Argentina. [View](#)

Visual Analytics for Linked Open Data in Marine Sciences. X Cloud Computing, Big Data & Emerging Topics, 2022. [View](#)

Mapyzer: una herramienta de carga y visualización de datos espacio-temporales. CACIC 2021. [View](#)

TEACHING EXPERIENCE

Teaching Assistant — Computer Science Fundamentals | UNPSJB | Feb 2025–Present
Supports first-year undergraduate teaching, practical activities, and student learning.

Teaching Assistant — Mathematical Analysis | UNPSJB | Aug 2021–Jul 2024

Student Tutor | UNPSJB | Apr 2019–Mar 2024
Provided academic guidance and supported student retention and university integration.

Computer Science Instructor | Fundación de Altos Estudios en Ciencias Comerciales | May–Aug 2022

Instructor — Mobile Technologies and Computing | UNPSJB | Sep–Dec 2019

SOFTWARE DEVELOPMENT EXPERIENCE

Software Developer | Cyberarg Sistemas | Mar 2023–Jul 2024
Contributed to the development and maintenance of production software systems.

Software Developer | CHESCA S.A. | Sep 2021–Jan 2022

Computer Laboratory Assistant | UNPSJB | Apr 2021–Mar 2023
Supported laboratory infrastructure, users, and academic computing activities.

Software Development Intern | UNPSJB | Sep 2020–Oct 2021

SELECTED POSTGRADUATE TRAINING

Year	Course	Institution
2026	Fundamentals of Explainable Artificial Intelligence	Universidad Nacional del Sur
2025	Pattern Recognition and Machine Learning	Universidad Nacional del Sur
2025	Image Processing and Computer Vision I	Universidad Nacional del Sur
2025	ADAPT Training Course on Ocean Best Practices	Invemar, Santa Marta, Colombia
2024	Deep Learning	Universidad Nacional de La Plata / CACIC

Year	Course	Institution
2024	Governance and Ethics in Information Management	Universidad Nacional del Sur
2023	Generative AI Fundamentals; Introduction to Responsible AI; Introduction to LLMs	Google Cloud Skills Boost

TECHNICAL SKILLS

AI & knowledge systems: Knowledge Graphs, GraphRAG/RAG, LLMs, neuro-symbolic reasoning, multi-agent systems, NLP, machine learning, deep learning

Programming & data: Python, JavaScript/TypeScript, R, SQL, SPARQL, RDF, OWL, GeoJSON

Frameworks & tools: Microsoft GraphRAG, Streamlit, NetworkX, Angular, Leaflet, Gephi, Git/GitHub

Scientific data: OBIS, GBIF, WoRMS, Linked Open Data, FAIR data, geospatial and marine biodiversity data

LANGUAGES

Spanish: Native • **English:** Upper-intermediate professional proficiency; ongoing training

ACADEMIC PROFILES

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